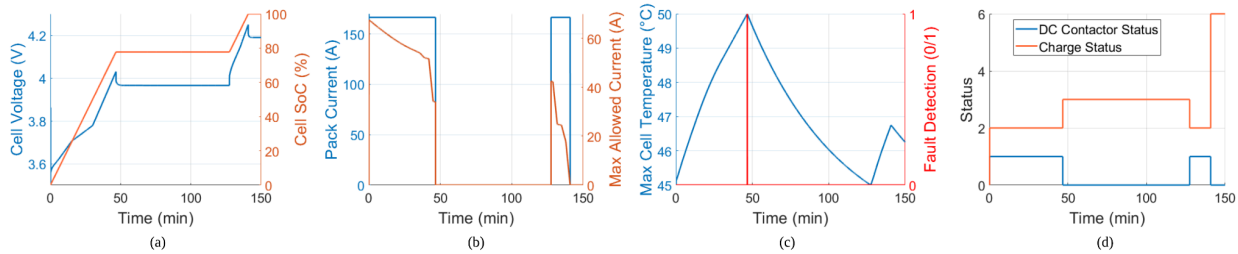

Variable Name	Plot Reference	Full Description (as provided)	Signal Source (Subsystem)
Cell Voltage	Subplot (a), blue line	Voltage measured through sensors, usually ranging from 2.5V to 4.2V for our particular Samsung INR21700-50E cells.	**Cell Monitoring Unit (CMU)**. It is an output of the CMU subsystem. CMU deals with cell-level measurements.
Cell SoC	Subplot (a), orange line	Cell State of Charge ranging between 0% and 100%. If the cell voltages are close to 4.15V, SoC = ~100% and if cell voltages are close to 3.0V, SoC = ~0%.	It is an output of the **SoC Estimation** subsystem.
Pack Current	Subplot (b), blue line	Flow of electricity into or out of the main terminals of the battery pack. If we want to charge our particular battery pack (105s34p) from 0% to 100% SoC in 1 hour, we need 166.6A because we have 34 cells in parallel (our battery pack config is 105s34p, which is 105 cells in series and 34 cells in parallel). Also, each cell has 4.9 AH of cell capacity. So, we need a continuous 4.9*34 = 166.6 A	**Battery Junction Box (BJB)**. It is an output of the BJB, which deals with pack-level measurements.

		to charge the battery pack from 0% to 100% in an hour.	
Max Allowed Charge Current	Subplot (b), orange line	We have a dataset (2D lookup table) that determines what the maximum allowed charge current would be for a particular pair of cell SoC and cell Temperature.	**Plug-In Charging** subsystem. Since the lookup table is located inside this subsystem, **Max Allowed Charge Current** is an output of PlugIn Charging subsystem.
Max Cell Temperature	Subplot (c), blue line	Maximum cell temperature of a particular parallel assembly of cells in the battery pack. We have 140 temperature sensors (4 sensors per module × 35 modules).	**Cell Monitoring Unit (CMU)**. It is an output of the CMU subsystem.
Fault Detection	Subplot (c), red line	A signal to indicate if any fault condition became true. If true, it will switch from 0 to 1. E.g., if max cell temperature >= 50°C, Fault Detection = 1 (an unsafe overheating condition).	**Plug-In Charging** subsystem. It is an output from this subsystem.
DC Contactor Status	Subplot (d), blue line	Indicates if the DC contactor is closed or open. 1 = Closed (charging ready), 0 = Open (charging can't happen), 2 = Faulted (e.g., stuck closed or stuck open).	**Plug-In Charging** subsystem. It is an output from this subsystem.
Battery Charge Status	Subplot (d), orange line	Indicates the present charge status. 0 = Not Ready, 1 = Ready, 2 = Charge Allow Current, 3 = Charge Disallow	**Plug-In Charging** subsystem. It is an output from this subsystem.

		Current, 6 = Complete, 7 = SNA.	